INCH-POUND

MIL-I-26689C(USAF)
AMENDMENT 2

1 JUNE 1994
SUPERSEDING
AMENDMENT 1
13 NOVEMBER 1987

MILITARY SPECIFICATION

INDICATOR, HORIZONTAL SITUATION,
AQU-2/A

This amendment forms a part of MIL-I-26689C(USAF), dated 10 June 1981, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.1 Line 1 add title "Scope" before the first sentence.

2.1 Delete title and first sentence and add: "Government documents."

Add: "2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

* Under SPECIFICATIONS, Federal, delete "PPP-B-601"; "PPP-B-636" entirely and add: "PPP-B-1672, Boxes Shipping, Reusable With Cushioning".

* Under SPECIFICATION, Military, change MIL-P-116's title to read "Preservation, Method Of. Delete "DOD-D-1000"; "MIL-E-5272" entirely and add: "MIL-B-121, Barrier Material, Greaseproofed, Flexible".

Lines 3 and 4 of the beneficial comments delete "HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016" and add: "Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990".

* Change ''FSC 6610'' to ''FSC 6605''.

AMSC N/A FSC 6610
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PAGE 2

* Under STANDARDS, Federal, add: "FED-STD-101 Preservation, Packaging And Packing Materials: Test Procedures".

* Under STANDARDS, Military, change "MIL-STD-143" to "MIL-STD-970". Delete "MIL-STD-826" entirely; add: "MIL-STD-461 Electromagnetic Interference Characteristics, Requirements For Equipment; "MIL-STD-462, Electromagnetic Interference Characteristics, Measurement Of"; "MIL-STD-810C Environmental Test Methods" and "MIL-STD-2073 DOD Material Procedures For Development And Application Of Packaging Requirements".

* Delete source statement and add: "(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)"

  Add: "2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

* Delete "3.1" entirely and add new: "3.1 Qualification. Indicator furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.3)."

PAGE 3

* 3.2 Line 3 delete "MIL-STD-143" and add: "MIL-STD-970".

PAGE 5

* 3.5 K add: "to isolated pins" at the end of the line.

* 3.5.1 Line 3 delete "MIL-STD-826" and add: "MIL-STD-461".

PAGE 7

3.9.2 Line 4 change to read "It shall be flat and free from discolorations (except as induced by glare reducing coatings), scratches and striae which interfere with the reading of the instrument."

* 3.9.4 Line 9 delete "is" and add: "it".

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3.9.4.2 Line 3 delete "100" and add: "10".

3.14.2 Line 3 delete "or" and add: "of".

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3.17.1 Line 10 delete "The course bar mechanism shall have a resistance of 1000 ohm ±3 p pins of figure 3 shall cause a deflection on the right when course arrow is on top. The course bar sensitivity shall be 75 ua per dot and range of displacement shall be ±2 dots."

PAGE 21

Delete "4.1" entirely and add: 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Add: "4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

* Delete "4.2"; "a", and "b" entirely and add: "4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (see 4.8.)."

* Delete "4.3" entirely and add: "4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in this specifications.

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* Add: "4.3.1 Retention of qualification. Each vendor listed on the Qualified Product List (QPL) will be required to certify their product on DD Form 1718 every two years."

* Delete "4.4" entirely and add: "4.4 Qualification inspection."

* 4.4.1 Delete "First article" in 2 places and add: Qualification inspection". Also delete "by the letter of authorization" at the end of the last sentence.

* 4.4.2 Delete "First article" and add: "Qualification inspection" in 3 places.

* 4.4.3 Delete "First article" and add: "qualification inspection" in 2 places.

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* 4.5.2.1 Line 3 delete "describer" and add: "described".

4.5.2.2,h Delete "spray" and add: "fog".

4.5.2.2,i Delete "Sand and dust" and add: "Dust (fine sand)."

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Under Group II Delete "spray" and add: "fog".

Under Group II Delete "sand and dust" and add: "dust (fine sand)."

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4.6.18 Delete existing paragraph and add: "Vibration error. The indicator shall be mounted, in its normal operating position, on a vibration fixture, and properly connected for operation. While being operated, the indicator shall be vibrated with circular motion in a plane inclined 45 degrees to the horizontal with a total vibration amplitude between 0.003 and 0.005 inches. The frequency of applied vibration shall be varied slowly from 5 to 50 cycles per second. While being vibrated, the indicator shall meet the following requirements."

PAGE 29

4.6.23 Line 1 delete "sample required in 3.26..1" and add: "cover glass and wedge."

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* 4.6.24 Line 2 delete "in accordance with MIL-STD-826" and add: "as required by MIL-STD-461, equipment category A 1b, and as described by MIL-STD-462".

4.6.25 Line 3 delete "MIL-E-5272" and add: "MIL-STD-810C."

4.6.25.1 Line 3 delete "procedures II" and add: "Method 501.1 Procedure I."

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4.6.25.2 Delete entire paragraph and add: "Temperature altitude. The indicator shall be subjected to a temperature altitude test in accordance with Method 504.1 of MIL-STD-810C for equipment category 6. There shall be no binding or sticking at any possible indication. After the indicator has returned to room temperature, it shall meet all individual tests. There shall be a 5 minute warm-up after the application of power before data is taken."

4.6.25.3 Line 2 add: Method 503.1" before "procedure I."

4.6.25.4 Line 2 add: "Method 507.1" before "procedure I."

4.6.25.6 Line 1 delete "salt spray and add: "salt fog" (2 places)

4.6.25.6 Line 2 add: "Method 509.1" before procedure I."

4.6.25.7 Line 1 delete "sand and dust" and add: "Dust (fine sand)" (2 places).

4.6.25.7 Line 2 add: "Method 510.1" before procedure I."

4.6.25.8 Line 2 delete "procedure V and add: "Method 514.2, procedure I, part 1 curve C and part 2 curve B."

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Delete "4.8" entirely.

* Change "4.8.1" to "4.8". Also change title to "Quality conformance."

* Delete Section 5 in its entirety and add:

"5 PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

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5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IC-1 In Accordance With (IAW) MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073 to provide a method IC-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073.

5.2 Packing. Packing shall be level "A", "B", or "C", as specified (see 6.2).

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit container shall be large enough to allow for the application of cushioning/wrap material, of sufficient density and thickness, to protect the unit container from punctures. Intermediate shipping containers shall be large enough to allow for the application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Fiberboard containers do not meet Level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedure and contracting activity.

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5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-129.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contracting activity and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101."

* Under 6. NOTES add: "(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)"

* Delete "6.3" entirely and add: "6.3 Qualification. "With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL) 26689 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is OC-ALC/TICLA, 3001 Staff Dr, Tinker AFB OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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